

ENGR 1451/2451: Exploratory Data Science

Expected Value and Standard Error of a Sample Mean

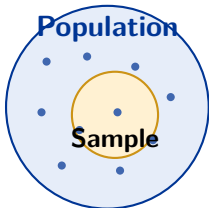
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Today: How Much Does a Sample Mean Vary?

- Distinguish **population parameters** from **sample statistics**.
- Find the **expected value** and **standard error of a sample mean**.
- Explain **chance error** and how sample size changes precision.
- Estimate the mean and its SE from **real data**.
- Introduce **Student's t** when the SE is estimated.

Point Estimates



Main idea

A **point estimate** is a single number from a sample used to estimate an unknown population parameter.

Statistic $\rightarrow$ Parameter

Quantity	Sample statistic	Population parameter
Mean	$\bar{x}$ or $\bar{y}$	μ
Standard deviation	s	σ
Variance	s^2	σ^2
Correlation	r	ρ
Regression intercept	b_0	β_0
Regression slope	b_1	β_1

- Statistics come from observed data.
- Population parameters are usually unknown.

Inference asks: how close is the statistic likely to be to the parameter?

Our Example: One Fair Six-Sided Die

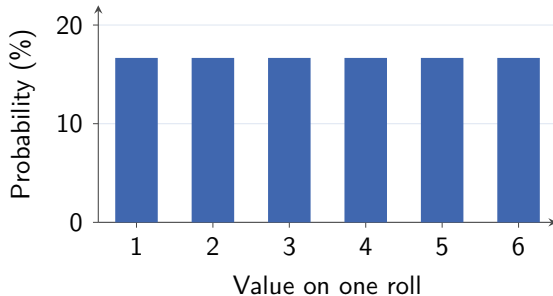
Population distribution

For one roll, the possible values are

1, 2, 3, 4, 5, 6.

Each has probability $1/6$.

- X is the value on one roll.
- Rolls are **independent**.
- A sample is a set of n rolls.



The Population Mean Is Known

Each of the six outcomes has probability $1/6$.

Average of the six equally likely values

$$\mu = E(X) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5.$$

The Population SD Is Also Known

The SD measures the spread of **individual roll values** around $\mu = 3.5$.

Value x	1	2	3	4	5	6
Deviation $x - 3.5$	-2.5	-1.5	-0.5	0.5	1.5	2.5
Squared deviation	6.25	2.25	0.25	0.25	2.25	6.25

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}, \quad N = 6.$$

$$\sigma = \sqrt{\frac{17.5}{6}} = \sqrt{\frac{35}{12}} \approx 1.708.$$

Population Parameters and Sample Statistics

Population: known for the fair die

$$\mu = 3.5, \quad \sigma \approx 1.708.$$

Sample: n observed rolls

Calculate $\bar{x}$ and s from the recorded values.

$$\bar{x} = \frac{x_1 + \cdots + x_n}{n}.$$

$\bar{X}$ varies across random samples; $\bar{x}$ is the mean observed in one sample.

Statistic estimates parameter

Quantity	Sample	Population
Mean	$\bar{x}$	μ
SD	s	σ
Variance	s^2	σ^2

A **point estimate** is one number from a sample used to estimate a population parameter.

Repeated Samples of 36 Rolls

Roll a fair six-sided die independently 36 times. Calculate the **sample mean** and repeat.

First 8 simulated samples

Sample	Mean $\bar{x}$	Error
1	3.3056	-0.1944
2	3.4167	-0.0833
3	3.3889	-0.1111
4	3.4722	-0.0278
5	2.9722	-0.5278
6	3.7500	+0.2500
7	3.1944	-0.3056
8	3.4722	-0.0278

$$\text{Error} = \bar{x} - 3.5$$

What stays the same?

$$E(\bar{X}) = 3.5$$

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{36}} \approx 0.285$$

A Sampling Distribution of Means

1. Roll the die 36 times.
2. Calculate the **sample mean**.
3. Collect 10,000 such means.

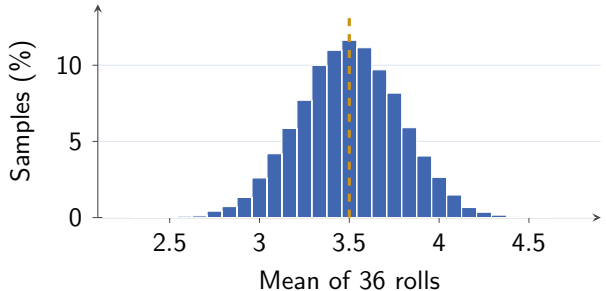
The histogram approximates the **sampling distribution of $\bar{X}$** .

10,000 simulated samples

Mean of means: **3.5**

SD of means: **0.2844**

Simulated SD of means: 0.2844. Theoretical SE: 0.2846.



Expected Value of the Sample Mean

For a random sample from a population with mean μ ,

The center of the sampling distribution

$$E(\bar{X}) = \mu.$$

For the fair die,

$$E(\bar{X}) = 3.5.$$

Standard Error of the Sample Mean

The **standard error** is the SD of the sample mean across repeated samples.

For n independent observations from the same population

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

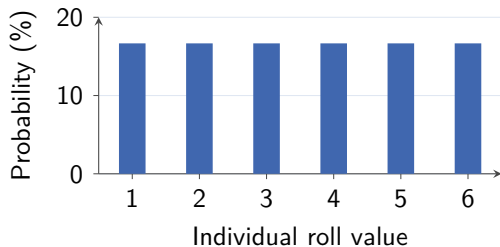
For 36 rolls of the fair die,

$$SE(\bar{X}) = \frac{\sqrt{35/12}}{\sqrt{36}} \approx \frac{1.708}{6} \approx 0.285.$$

SD and SE Describe Different Spreads

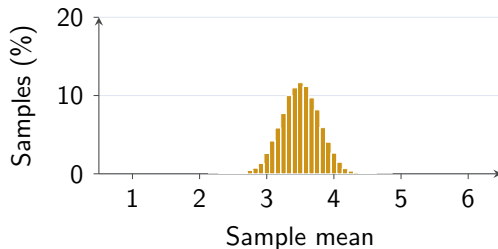
One roll

Population SD $\sigma \approx 1.708$



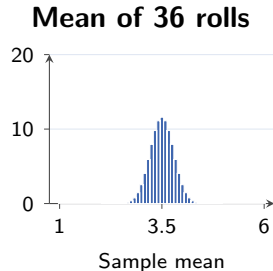
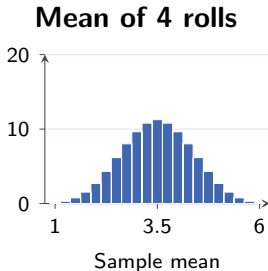
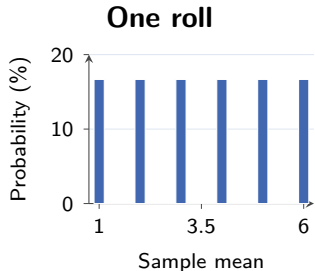
Mean of 36 rolls

Standard error $SE(\bar{X}) \approx 0.285$



Why a Normal Approximation Can Work

For independent rolls of this fair die, the distribution of the sample mean becomes approximately normal as n grows. This illustrates the **central limit theorem**.



A Probability for the Sample Mean

For 36 rolls, $E(\bar{X}) = 3.5$ and $SE(\bar{X}) \approx 0.2846$.

How often is $\bar{X}$ between **3** and **4**?

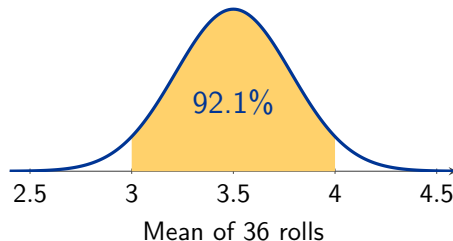
A z-score counts SEs from 3.5.

$$z_{\text{lower}} = \frac{3 - 3.5}{0.2846} \approx -1.76$$

$$z_{\text{upper}} = \frac{4 - 3.5}{0.2846} \approx 1.76$$

Normal approximation

$$P(3 \leq \bar{X} \leq 4) \approx 92.1\%.$$



When Population Parameters Are Unknown

With real measurements, we usually observe **a sample** and estimate the population quantities.

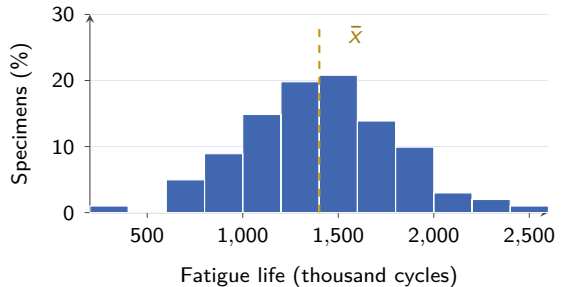
Quantity	Population quantity	Estimate from data
Population mean	μ	Sample mean $\bar{x}$
Population SD	σ	Sample SD s
SE of the sample mean	$\sigma/\sqrt{n}$	$s/\sqrt{n}$

Real Data: Fatigue Life of Aluminum

- **101** rectangular specimens of 6061-T6 aluminum.
- Repeated loading; maximum stress **21,000 psi**.
- Measurement: **thousands of cycles until rupture**.

Population of interest

Comparable specimens manufactured and tested under the **same conditions**.



Data: NIST/SEMATECH e-Handbook, Fatigue Life of Aluminum Alloy Specimens.

Estimate the Population Mean and SD

Use all **101 measured fatigue lives**. The population parameters remain unknown.

Estimate μ with the sample mean

$$\bar{x} = \frac{x_1 + \cdots + x_n}{n}$$

$$\bar{x} \approx 1400.9$$

Estimate σ with the sample SD

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

$$s \approx 391.3$$

Both estimates are in **thousand cycles**.

Use $n - 1$ for sample variance because the population mean was estimated from the same sample.

Estimate the Standard Error of the Mean

For independent, comparable specimens, substitute s for the unknown σ .

Estimated standard error

$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{391.3}{\sqrt{101}} \approx 38.9 \text{ thousand cycles.}$$

- **Sample SD: 391.3 thousand cycles.** Spread of individual lifetimes.
- **Estimated SE: 38.9 thousand cycles.** Estimated spread of means from samples of 101 specimens.

When the SE Is Estimated: Student's t

With real data, $s/\sqrt{n}$ is an **estimate** of the standard error.

Both the sample mean and sample SD vary

For independent observations from a normal population,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

S is the sample SD; its observed value is s .

- The t curve is symmetric about 0, with **heavier tails** than the standard normal.
- Its shape depends on the **degrees of freedom**: $n - 1$.
- As n grows, the t curve approaches the standard normal.

What Can We Know About This Sample?

Fair die: known population

- True mean: $\mu = 3.5$.
- Known SE for 36 rolls: **0.285**.

Fatigue data: unknown population

- True mean μ : **unknown**.
- Observed mean: $\bar{x} = 1400.9$ thousand cycles.
- Actual chance error $\bar{x} - \mu$: **unknown**.
- Estimated SE: **38.9** thousand cycles.

Quick Check

1. For the same fair die, what is the expected mean of **144 rolls**?
2. Moving from 36 to 144 rolls, what happens to the **SE of the mean**? What happens to the SD of individual rolls?
3. In the fatigue dataset, which number describes the observed spread of individual lifetimes: **391.3** or **38.9**?
4. Can we calculate the actual chance error of the fatigue sample mean? Why or why not?

Summary

- **Population SD** σ describes individual-value spread.
- **SE of the mean** describes the spread of sample means.

Independent observations from the same population

$$E(\bar{X}) = \mu, \quad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}, \quad \widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

- **Fair die:** $\mu = 3.5$ and $\sigma \approx 1.708$ are known; 36 rolls give $SE(\bar{X}) \approx 0.285$.
- **Real data:** $\bar{x}$ estimates μ , s estimates σ , and $s/\sqrt{n}$ estimates the SE.
- **Student's t** describes the mean standardized by an estimated SE under the stated sampling assumptions.